

Performance Analysis and Code Design for Resistive Random-Access Memory Using Channel Decomposition Approach

Guanghui Song[✉], *Member, IEEE*, Meiru Gao, Ying Li[✉], *Member, IEEE*, Bin Dai[✉], *Member, IEEE*, Kui Cai[✉], *Senior Member, IEEE*, and Lin Zhou[✉], *Senior Member, IEEE*

Abstract—An analytical framework integrating performance characterization and coding theory is proposed to mitigate sneak path (SP) interference in resistive random-access memory (ReRAM) crossbar arrays. The core innovation is identified in the mathematical decomposition of ReRAM's non-ergodic data-dependent channel into multiple stationary memoryless subchannels. Through information-theoretic analysis, an approximate finite-length characterization of the theoretical lower bound for decoding word error probability (WEP) is established. This is achieved by systematically analyzing the SP occurrence rate in constrained array geometries combined with comprehensive evaluation of both mutual information and dispersion metrics across the decomposed channel components. Building upon this decomposition paradigm, a systematic code construction methodology is developed using density evolution principles for sparse-graph code design. The designed codes not only exhibit capacity-approaching decoding thresholds but also yield word error rate simulation results that are close to the derived WEP bound under practical crossbar configurations.

Index Terms—Non-volatile memory (NVM), resistive random-access memory (ReRAM), sneak path, channel decomposition, code design.

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Guanghui Song, Meiru Gao, and Ying Li are with the School of Communication Engineering, Xidian University, Xi'an 710071, China (e-mail: songguanghui@xidian.edu.cn; 1132079534@qq.com; yli@mail.xidian.edu.cn).

Bin Dai is with the School of Internet of Things, Nanjing University of Posts and Telecommunications, Nanjing 210094, China, and also with the Science, Mathematics and Technology Cluster, Singapore University of Technology and Design (SUTD), Singapore 487372 (e-mail: daibin@njupt.edu.cn).

Kui Cai is with the Science, Mathematics and Technology Cluster, Singapore University of Technology and Design, Singapore 487372 (e-mail: cai_kui@sutd.edu.sg).

Lin Zhou is with Xiamen Key Laboratory of Mobile Multimedia Communications, College of Information Science and Engineering, Huaqiao University, Xiamen 361021, China (e-mail: linzhou@hqu.edu.cn).

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I. INTRODUCTION

RESISTIVE random-access memory (ReRAM) is a promising emerging non-volatile memory that utilizes memory cells arranged typically in a crossbar array to store information [1], [2], [3], [4]. Each memory cell is essentially a memristor with two distinct resistance states, i.e., the High-Resistance State (HRS) with a resistance value of R_0 and the Low-Resistance State (LRS) with a resistance value of R_1 . These states can be used to represent one bit of information, where HRS corresponds to a bit value of 0 and LRS to a bit value of 1. Consequently, an $N \times N$ memory array comprising N^2 memory cells is sufficient for storing an $N \times N$ binary data array $x^{N \times N} = [x_{m,n}]_{N \times N}$, where $x_{m,n}$ is either 0 or 1.

Sneak path (SP) is an undesired current path during memory reading which severely degrades the data detection performance, [5], [6], [7], [8], [9], as illustrated in Fig. 1. When a memory cell is read, a voltage is applied across the corresponding word line and bit line, causing current to flow through the cell. An SP represents an alternative current path that runs parallel to the intended path for resistance measurement. This alternative path forms a closed loop, originating and terminating at the target cell being read, while traversing through LRS cells via alternating vertical and horizontal steps. Fig. 1 depicts an SP during the reading of cell (3, 2). The green line indicates the desired path for resistance measurement, while the red line, passing through cells (3, 4) $\rightarrow$ (1, 4) $\rightarrow$ (1, 2), represents an SP. In such scenarios, the detected resistance value (in the absence of noise) becomes

$$R'_0 = \left(\frac{1}{R_0} + \frac{1}{R_s} \right)^{-1} \quad (1)$$

where R_s represents the parasitic resistance introduced by the SP. Typically, we have $R_0 \gg R'_0 > R_1$. Since the SP always decreases the measured resistance value, it is harmful only when a cell with HRS is read. Fig. 2 depicts the probability density function (PDF) of the readback signal $y_{m,n}$, which represents the detected resistance value contaminated with Gaussian noise, for memory cell (m, n) . This illustration considers scenarios where the stored data is either $x_{m,n} = 1$ or $x_{m,n} = 0$, with and without the interference of an SP. We adhere to the parameter setting employed in previous works [10], [11], [12], [13], where the respective resistance values are designated as $R_0 = 1000 \Omega$, $R_1 = 100 \Omega$, and $R_s = 250 \Omega$. Notably, a significant cross-error probability is observed in the presence of an SP. In the following, cell (m, n) is referred to as an LRS cell if $x_{m,n} = 1$; as an HRS cell if $x_{m,n} = 0$ and the

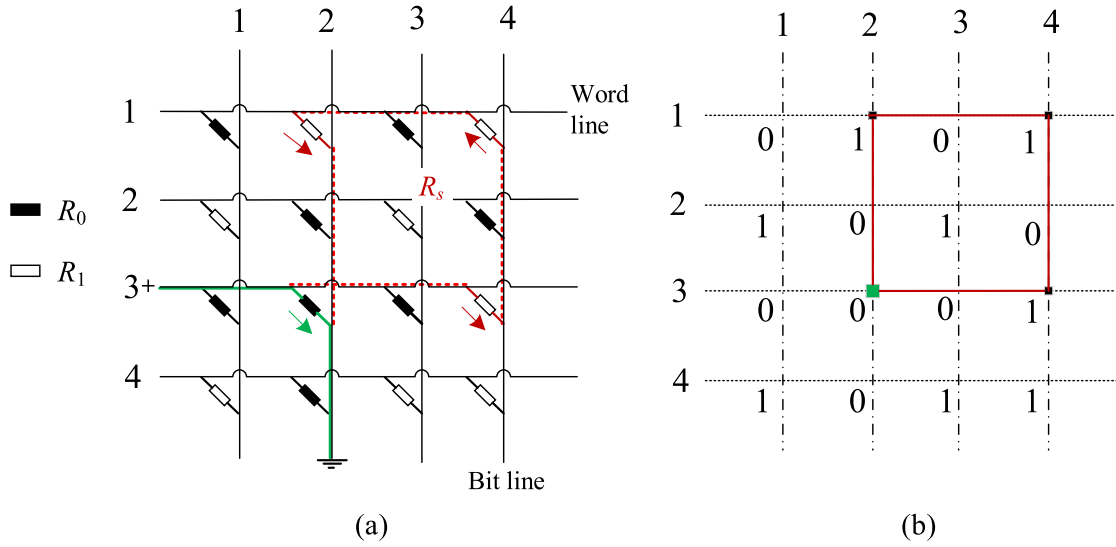


Fig. 1. (a) A 4×4 ReRAM array and (b) The corresponding data array. The green line plots the desired path when measuring the resistance of cell (3, 2), and the red line forms an SP. Arrows show current flow directions. A reverse current flows across cell (1, 4).

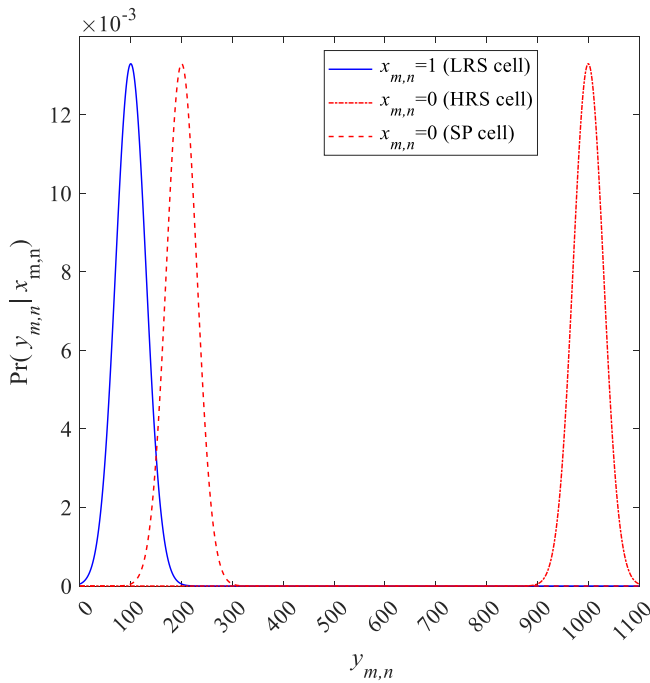


Fig. 2. PDF of the detected resistance value $y_{m,n}$ when the cell is an LRS cell ($x_{m,n} = 1$), a HRS cell ($x_{m,n} = 0$ without SP interference), or an SP cell ($x_{m,n} = 0$ with SP interference). The associated resistance values are $R_0 = 1000 \, \Omega$, $R_1 = 100 \, \Omega$, and $R_s = 250 \, \Omega$, and the noise standard deviation is $\sigma = 30$.

cell is not affected by an SP; and as an SP cell if $x_{m,n} = 0$ and the cell is affected by an SP.

The most commonly adopted method to mitigate SPs is the utilization of cell selectors, which are electrical devices that permit current flow in only one direction across memory cells. One example of this is the one-diode one-resistor (1D1R) type ReRAM, which incorporates a diode in series with each memory cell. Since an SP inherently generates a reverse

current in at least one of the cells (e.g. cell (1, 4) in Fig. 1) along the SP, if all selectors are functioning correctly, SPs can be completely prevented. However, selectors are also prone to errors due to imperfections in the production or maintenance of the memory leading to the reoccurrences of SPs, as reported in [10], [11], [12], and [13]. Therefore, (m, n) is referred to as an SP cell if and only if the following three conditions (*SP conditions*) are met:

- 1) $x_{m,n} = 0$.
- 2) There exists $i, j \in \{1, \dots, N\}$ that $x_{m,j} = x_{i,j} = x_{i,n} = 1$.
- 3) The selector located at cell (i, j) is faulty (a reverse current flowing through this cell).

Based on previous works [11], [12], [13], the occurrence of selector failure (SF) is random and is unknown to the data detector. According to conditions 2) and 3), an SF at position (i, j) can only result in SPs if the cell at this position is an LRS cell, meaning $x_{i,j} = 1$. Otherwise, the failure will have no impact on the array. Therefore, an SF occurring at an LRS cell is referred to as an *active SF*. Furthermore, the SP conditions above specifically constrain SPs of length 3, traversing exactly three LRS cells. Longer SPs are ignored due to their insignificant impact compared to those of length 3. Consistent with previous works [11], [12], [13], we do not take into account the superposition effect of multiple SPs. A more sophisticated SP model was explored in [10].

Recent works [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20] addressed the SP problem using information theory and channel coding methods. For uncoded ReRAM, works [10], [11], [12] proposed efficient data detection schemes. Ben-Hur and Cassuto [10] presented a cell-independent data detection scheme without considering the inter-cell correlation. Chen et al. [11] and our previous work [12] proposed joint data and SP detection schemes. In [12], we first detected the locations of SFs, based on which, the SPs and data were jointly detected. With an assumption of

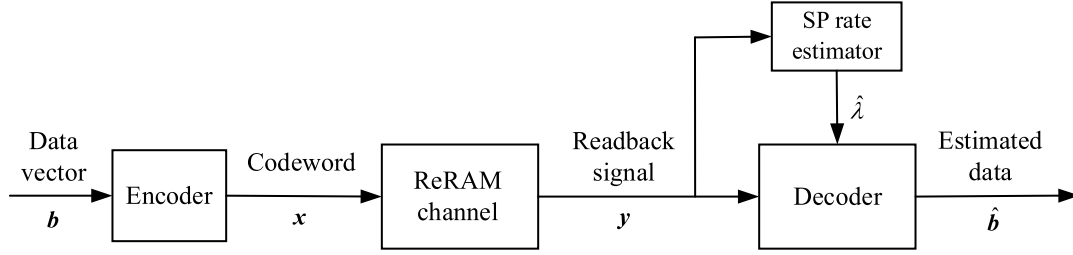


Fig. 3. System model of Coded ReRAM.

less than 3 SFs in the array, our scheme yields a near-optimal performance.

To further enhance the data storage reliability, channel codes including low-density parity-check (LDPC) and polar codes were applied to ReRAM [13], [14], [15], [16], [17]. Regarding the code design, a significant challenge arises from the non-ergodic nature of the ReRAM channel as shown in [20]. In this scenario, due to the data dependency of SPs, the channel's performance fluctuates for each data array, especially when the array size is small. The channel was characterized as time-varying in [13]. However, most previous works have overlooked this factor during code construction, instead designing the code based on an averaged channel performance.

This work presents an analytical framework unifying performance characterization and coding theory for ReRAM systems. The foundational innovation resides in the mathematical decomposition of ReRAM's non-ergodic data-dependent channel into multiple stationary memoryless subchannels. Through information-theoretic analysis, we derive an approximate finite-length characterization of the theoretical lower bound for decoding word error probability (WEP), thereby creating theoretically grounded benchmarks for code design. This is achieved by systematically analyzing SP occurrence rates in geometrically constrained arrays combined with comprehensive evaluation of mutual information and dispersion metrics across decomposed channel components. The proposed decomposition methodology provides dual advantages: (i) systematic characterization of non-ergodic channel dynamics, and (ii) computationally tractable code construction pathways. Building upon this theoretical framework, we develop a practical code design architecture employing density evolution (DE)-optimized sparse graph codes. Key to this implementation is the establishment of Gaussianity in log-likelihood ratio (LLR) message distributions. This property enables the derivation of low-complexity two-dimensional DE equations governing code optimization. The designed codes not only exhibit capacity-approaching decoding thresholds but also yield word error rate (WER) simulation results that are close to the derived WEP bound under practical crossbar configurations.

The rest of this paper is organized as follows. Sec. II introduces the coded ReRAM system model. In Sec. III, we propose the channel decomposition and conduct the finite-length performance analysis. Sec. IV presents the sparse-graph code design and develops the two-dimensional DE framework. Numerical results of code optimization and simu-

lations are presented in Sec. V. Finally, we conclude our work in Section VI.

II. CODED RERAM SYSTEM MODEL

The system model of coded ReRAM is illustrated in Fig. 3. In this model, data vector $\mathbf{b} = (b_1, b_2, \dots, b_{N^2R})$ is encoded into an $N \times N$ codeword array $\mathbf{x} = [x_{m,n}]_{N \times N}$ where $x_{m,n} \in \{0, 1\}$ and R represents the code rate. This encoded array $\mathbf{x}$ can then be stored in an $N \times N$ ReRAM array. Define

$$R_x(e) \triangleq \left(\frac{1}{R_x} + (1-x)\frac{e}{R_s} \right)^{-1}, \quad \text{for } x, e \in \{0, 1\} \quad (2)$$

where x denotes the data bit value and e is a binary indicator of whether an SP occurred. Thus, $R_1(0) = R_1(1) = R_1$ and $R_0(0) = R_0, R_0(1) = R'_0$. Let Ψ be the set that includes all the indices of cells with SFs in the array. According to the SP conditions, the readback signal $\mathbf{y} = [y_{m,n}]_{N \times N}$ is written as

$$y_{m,n} = R_{x_{m,n}} \left(\bigcup_{(i,j) \in \Psi} x_{m,j} x_{i,j} x_{i,n} \right) + z_{m,n}, \quad (3)$$

where $\bigcup$ is the logical OR operator, i.e., $\bigcup_{(i,j) \in \Psi} x_{m,j} x_{i,j} x_{i,n} = 1$ if at least one of $(i, j) \in \Psi$ with $x_{m,j} x_{i,j} x_{i,n} = 1$, otherwise, $\bigcup_{(i,j) \in \Psi} x_{m,j} x_{i,j} x_{i,n} = 0$, and $z_{m,n} \sim \mathcal{N}(0, \sigma^2)$, $m = 1, \dots, N, n = 1, \dots, N$ are independent and identically distributed (i.i.d.) samples of Gaussian noise.

An LLR for each coded bit is calculated to facilitate the soft decoding. Different from the conventional binary input Gaussian channel, where the data bits 0 and 1 are modulated to two distinct voltage values, in (3), if $x_{m,n} = 0$, the resistance value of cell (m, n) may vary as R_0 or R'_0 depending on whether the cell is affected by an SP. Assuming Bernoulli (q) data storage where $\Pr(x_{m,n} = 1) = q$ and $\Pr(x_{m,n} = 0) = 1 - q$ with $0 < q < 1$, the LLR for $x_{m,n}$ is

$$\begin{aligned} L(x_{m,n}|y_{m,n}) &= \log \frac{(1-q) \Pr(y_{m,n}|x_{m,n}=0)}{q \Pr(y_{m,n}|x_{m,n}=1)} \\ &= \log \frac{\lambda \phi(y_{m,n} - R'_0) + (1-\lambda) \phi(y_{m,n} - R_0)}{\phi(y_{m,n} - R_1)} + \log \frac{1-q}{q} \end{aligned} \quad (4)$$

where $\lambda, 0 \leq \lambda \leq 1$, is the probability that cell (m, n) is an SP cell, given that $x_{m,n} = 0$, and $\phi(z) \triangleq 1/(\sqrt{2\pi}\sigma)e^{-\frac{z^2}{2\sigma^2}}$ is a Gaussian PDF with mean 0 and variance σ^2 .

Strictly speaking, λ is cell-specific and its value is related to the data pattern of the whole array $\mathbf{x}$. Thus, the optimal decoder should process the memory cells jointly, but the complexity will be very high [12]. A simpler decoder that

facilitates practical realization is carried out by treating the memory cell independently [13], [14], [15], [16], [17]. In this scenario, λ is assumed to be array-specific but within the same array, a common λ is adopted for all memory cells. The parameter λ is referred to as the *SP rate* of the array, reflecting the fraction of SP cells among the cells that are intended to store the data 0. Obviously, the SP rate depends on both the number of active SFs and the data pattern of the array. In practice, λ is first inferred based on the readback signal array before decoding. Following previous works [13], [14], [15], [16], [17], λ is estimated based on the hard decision of the readback signals. Specifically, let

$$\bar{y}_{m,n} = \arg \min_{r \in \{R'_0, R_0, R_1\}} |y_{m,n} - r|.$$

The estimation of λ is given by

$$\hat{\lambda} = \frac{n_{R'_0}}{n_{R'_0} + n_{R_0}}, \quad n_r \triangleq \sum_{m=1}^N \sum_{n=1}^N 1\{\bar{y}_{m,n} = r\} \quad (5)$$

where $1\{\mathcal{E}\}$ is the indication function which takes the value $1\{\mathcal{E}\} = 1$ if $\mathcal{E}$ is satisfied; otherwise, it takes value $1\{\mathcal{E}\} = 0$.

III. CHANNEL DECOMPOSITION AND FINITE-LENGTH ANALYSIS

The SP rate affects the channel quality significantly. Furthermore, it depends on the number of active SFs in the array and the data pattern, both of which are random. This poses a significant challenge to code design, as channel codes are typically optimized for stationary memoryless channels. Designing a code based on the average channel SP rate risks high decoding failure probability when the actual SP rate surpasses this average. On the contrary, designing a code based on the worst-case scenario results in a notably low code rate. In this section, we propose a channel decomposition approach that decomposes the ReRAM channel into multiple stationary memoryless channels, each with a distinct SP rate. By designing a code tailored to one of these stationary memoryless channels, we can achieve a specific rate with an outage probability. We analyze the probability distribution of the SP rate and the second-order rate of the sub-channels for a finite length of N and derive an approximate finite-length performance bound. This bound serves as a benchmark for code design.

We define a λ -Gaussian channel as a stationary and memoryless channel characterized by binary input and continuous output, as illustrated in Fig. 4. This λ -Gaussian channel was also termed the (λ, σ) -channel in [13]. When $\lambda = 0$ or $\lambda = 1$, the λ -Gaussian channel reduces to a binary-input additive white Gaussian noise (BIAWGN) channel.

For a given array size of $N \times N$, if the data input distribution and the SF probability distribution jointly yield a probability distribution of the SP rate, denoted as $F_N(\lambda)$, $0 \leq \lambda \leq 1$, we say that the ReRAM channel can be decomposed into a set of λ -Gaussian channels with a profile of $F_N(\lambda)$. In other words, the ReRAM channel is a mixture of multiple λ -Gaussian channels, where $F_N(\lambda)$ specifies the proportion of each constituent channel within the mixture. Let I_λ and V_λ be the mutual information and dispersion of the λ -Gaussian channel.

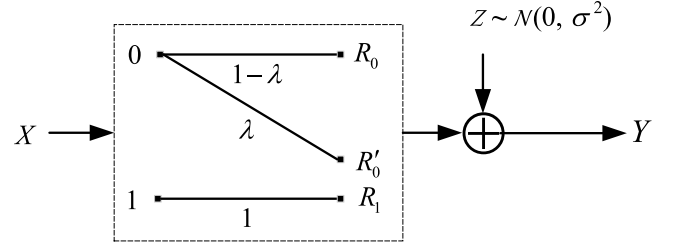


Fig. 4. λ -Gaussian channel model.

According to the Polyanskiy-Poor-Verdu (PPV) bound [21], the minimum WEP achievable using a code with rate R and length N^2 over the ReRAM channel can be approximated by

$$P_{\text{error}} \approx \int_{\lambda} F_N(\lambda) Q\left(\frac{N(I_\lambda - R)}{\sqrt{V_\lambda}}\right) d\lambda \quad (6)$$

where $Q(x) = \int_x^\infty 1/\sqrt{2\pi} e^{-\frac{u^2}{2}} du$. Notably, the expression $Q\left(\frac{N(I_\lambda - R)}{\sqrt{V_\lambda}}\right)$ approximates the minimum WEP for codes over λ -Gaussian channel, derived through neglecting a higher-order term [21]. The proportion of this higher-order term in the total WEP is less than $O\left(\frac{\log N}{N^2}\right)$. Consequently, the right-hand side (RHS) of (6) represents an approximation of the theoretical lower bound on WEP for ReRAM channels.

In the rest of this section, we provide approximation analyses for $F_N(\lambda)$, I_λ , and V_λ . In our analyses, we assume Bernoulli (q) data storage and with a fixed number of K ($K \ll N$) active SFs in the array. Under this assumption, the distribution of λ depends solely on the randomness of the data pattern. However, this does not affect the generality of our analyses. For the scenarios involving a random number of SFs, we can first average $F_N(\lambda)$ over the probability distribution of the SF number and then apply the bound given in (6).

A. Probability Distribution of SP Rate $F_N(\lambda)$

Considering Bernoulli (q) data storage with K active SFs in an $N \times N$ ReRAM array, we analyze the probability distribution of the SP rate λ_N . We assume a type of scattered SF pattern where the SFs are located in different rows and columns as shown in Fig. 5. The analysis can be extended to the case of a non-scattered SF pattern. We first show the convergence of SP rate as $N \rightarrow \infty$ by analyzing its mean and variance. An earlier version of this result appeared in our conference paper [22]. Then we propose a Gaussian approximation for $F_N(\lambda)$.

We show that if the data is i.i.d. Bernoulli (q), as $N \rightarrow \infty$, with probability 1,

$$\lambda_N \rightarrow 1 - \left(1 - \frac{2K}{N}\right) (1-q^2)^K - \frac{2K}{N} (1-q^2)^{K-1} + o\left(\frac{1}{N}\right) \quad (7)$$

where we neglected higher-order terms of $o(1/N)$ but kept the $O(1/N)$ terms to ensure our analysis remains valid for finite but sufficiently large array sizes. In the following, we demonstrate that the expectation of the SP rate, $E[\lambda_N]$, tends to the RHS

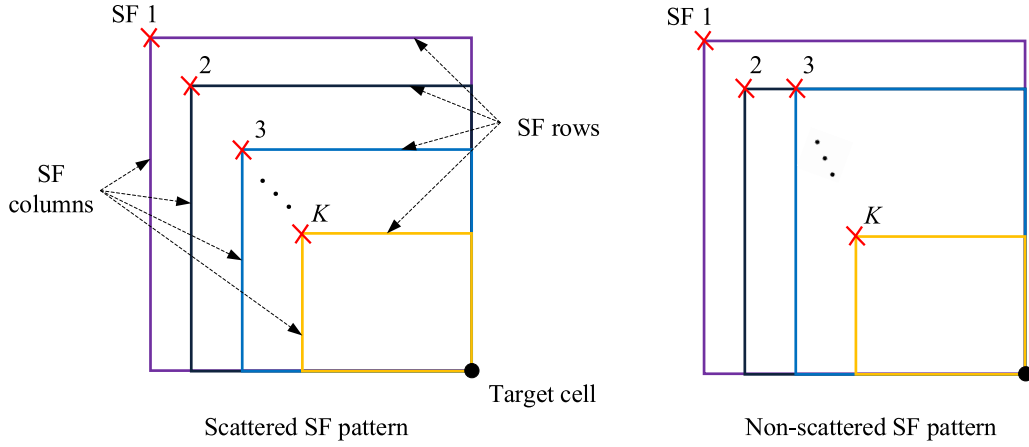


Fig. 5. An example of scattered and non-scattered SF patterns.

of (7), and its variance $D[\lambda_N] \rightarrow O(1/N)$ as $N \rightarrow \infty$. Thus, given $\epsilon > 0$, Chebyshev's inequality yields

$$\Pr(|\lambda_N - E[\lambda_N]| \geq \epsilon) \leq \frac{D[\lambda_N]}{\epsilon^2} \rightarrow 0.$$

1) *Expectation $E[\lambda_N]$* : Let X_{sf} be the set containing the indices of memory cells that belong to the SF rows and columns and X_{nsf} be the set containing the indices of the remaining memory cells. Here an SF row or column refers to the row or column that contains an active SF, as illustrated in Fig. 5. We divide the data array $\mathbf{x}$ into two parts: $\{x_{m,n}, (m,n) \in X_{sf}\}$ and $\{x_{m,n}, (m,n) \in X_{nsf}\}$. For ReRAM with K scattered active SFs, the cardinality of these two sets are

$$\begin{aligned} |X_{sf}| &= 2KN - K^2 \\ |X_{nsf}| &= N^2 - 2KN + K^2. \end{aligned}$$

From (3), cell (m,n) will be affected by SP if and only if $x_{m,n} = 0$ and $\bigcup_{(i,j) \in \Psi^*} x_{m,j}x_{i,n} = 1$, where Ψ^* is the set containing the indices of all the active SFs. The SP rate λ_N of an array is the total number of memory cells affected by SPs, normalized by the factor $(1-q)N^2$. Here $(1-q)N^2$ represents the expected number of cells containing data 0. Thus, λ_N is derived as

$$\begin{aligned} \lambda_N &= \frac{1}{(1-q)N^2} \sum_{m=1}^N \sum_{n=1}^N 1 \left\{ \bigcup_{(i,j) \in \Psi^*} x_{m,j}x_{i,n} = 1, x_{m,n} = 0 \right\} \\ &= \frac{1}{(1-q)N^2} \sum_{(m,n) \in X_{nsf}} 1 \left\{ \bigcup_{(i,j) \in \Psi^*} x_{m,j}x_{i,n} = 1, x_{m,n} = 0 \right\} \\ &\quad + \frac{1}{(1-q)N^2} \sum_{\substack{(m,n) \in X_{sf} \\ (m,n) \notin \Psi^*}} 1 \left\{ \bigcup_{\substack{(i,j) \in \Psi^* \\ i \neq m, j \neq n}} x_{m,j}x_{i,n} = 1, x_{m,n} = 0 \right\} \end{aligned}$$

where the second term of the above expression can be explained as follows: first $(m,n) \in \Psi^*$ is in conflict with the condition $x_{m,n} = 0$; second, if either $i = m$ or $j = n$, the product $x_{m,j}x_{i,n} = 1$ would contradict the condition $x_{m,n} = 0$. Using $1\{\bigcup_j x_j = 1, x = 0\} = (1 - \prod_j (1 - x_j))(1 - x)$, after

neglecting the high order terms of $o(1/N)$, the above equation is rewritten as

$$\lambda_N \rightarrow \left(1 - \frac{2K}{N}\right) A_N + \frac{2K}{N} B_N + o\left(\frac{1}{N}\right) \quad (8)$$

where

$$\begin{aligned} A_N &= \frac{1}{|X_{nsf}|(1-q)} \\ &\quad \sum_{(m,n) \in X_{nsf}} (1 - x_{m,n}) \left(1 - \prod_{(i,j) \in \Psi^*} (1 - x_{m,j}x_{i,n})\right) \\ B_N &= \frac{1}{(|X_{sf}| - K)(1-q)} \\ &\quad \sum_{\substack{(m,n) \in X_{sf} \\ (m,n) \notin \Psi^*}} (1 - x_{m,n}) \left(1 - \prod_{\substack{(i,j) \in \Psi^* \\ i \neq m, j \neq n}} (1 - x_{m,j}x_{i,n})\right). \end{aligned}$$

Using the i.i.d. assumption of data, we have

$$\begin{aligned} E[A_N] &= \frac{1}{|X_{nsf}|(1-q)} \sum_{(m,n) \in X_{nsf}} (1 - E[x_{m,n}]) \\ &\quad \times \left(1 - \prod_{(i,j) \in \Psi^*} (1 - E[x_{m,j}]E[x_{i,n}])\right) \\ &= 1 - (1 - q^2)^K \end{aligned} \quad (9)$$

and $E[B_N] \rightarrow 1 - (1 - q^2)^{K-1}$, where we used $|\Psi^*| = K$. Therefore,

$$\begin{aligned} E[\lambda_N] &\rightarrow 1 - \left(1 - \frac{2K}{N}\right) (1 - q^2)^K - \frac{2K}{N} (1 - q^2)^{K-1} \\ &\quad + o\left(\frac{1}{N}\right). \end{aligned} \quad (10)$$

Note that the scattered SF pattern represents the worst-case scenario. For the non-scattered SF pattern shown in Fig. 5 (right), the expected SP rate, when neglecting the $o(1)$ terms, will be $1 - (1 - 2q^2 + q^3)(1 - q^2)^{K-2}$. This value is smaller than the RHS of (9). For a more general case, we define a type- k SF sub-pattern as one consisting of k SFs (where $1 \leq k \leq K$)

aligned in a single row or column. The overall SF pattern can then be characterized by a type-vector $(n_1, n_2, \dots, n_K)$, where n_k denotes the number of type- k sub-patterns present in the K -SF pattern. Note that the scattered SF pattern is a special case corresponding to the type-vector $(K, 0, \dots, 0)$, while the non-scattered pattern shown in Fig. 5 (right) has the type-vector $(K-2, 1, 0, \dots, 0)$. By taking the expectation of A_N , we derive the following general expression for the asymptotic expectation of the SP rate

$$1 - \prod_{k=1}^K (1 - q + q(1-q)^{n_k}).$$

It should be emphasized that if an SF shares both a row and a column with other SFs, it does not contribute to the asymptotic expectation of the SP rate and must be excluded in advance.

2) *Variance $D[\lambda]$* : Based on (8), we have

$$D[\lambda_N] \rightarrow \left(1 - \frac{2K}{N}\right)^2 D[A_N] + \left(\frac{2K}{N}\right)^2 D[B_N] + \frac{4K}{N} \left(1 - \frac{2K}{N}\right) E[(A_N - E[A_N])(B_N - E[B_N])]. \quad (11)$$

We analyze $D[A_N]$ and then show that the remaining terms are all $o(1/N)$.

Once again, utilizing the i.i.d. assumption of data, we have

$$\begin{aligned} E[A_N^2] &= \frac{1}{|X_{\text{nsf}}|^2(1-q)^2} \sum_{(m,n) \in X_{\text{nsf}}} \sum_{(m',n') \in X_{\text{nsf}}} E[(1-x_{m,n})(1-x_{m',n'})] \\ &\times E\left[\left(1 - \prod_{(i,j) \in \Psi^*} (1-x_{m,j}x_{i,n})\right)\right. \\ &\left.\left(1 - \prod_{(i,j) \in \Psi^*} (1-x_{m',j}x_{i,n'})\right)\right]. \end{aligned}$$

We simplify the above expression as follows. First, the value inside the first summation of $\sum_{(m,n) \in X_{\text{nsf}}}$ is independent of the index (m,n) . Second, the second expectation can be further expanded as

$$\begin{aligned} &E\left[\left(1 - \prod_{(i,j) \in \Psi^*} (1-x_{m,j}x_{i,n})\right)\right. \\ &\left.\left(1 - \prod_{(i,j) \in \Psi^*} (1-x_{m',j}x_{i,n'})\right)\right] \\ &= 1 - 2(1-q^2)^K + (E_{m',n'})^K \end{aligned}$$

where

$$E_{m',n'} = E[(1-x_{m,j}x_{i,n})(1-x_{m',j}x_{i,n'})].$$

Thus, we have the following simplified expression

$$\begin{aligned} E[A_N^2] &= \frac{1}{|X_{\text{nsf}}|(1-q)^2} \sum_{(m',n') \in X_{\text{nsf}}} E[(1-x_{m,n})(1-x_{m',n'})] \\ &\times \left(1 - 2(1-q^2)^K + (E_{m',n'})^K\right). \end{aligned}$$

$$\times \left(1 - 2(1-q^2)^K + (E_{m',n'})^K\right).$$

For $(m,n) \in X_{\text{nsf}}$, we partition the set X_{nsf} into four subsets:

$$\begin{aligned} S_1 &= \{(m',n') | (m',n') = (m,n)\} \\ S_2 &= \{(m',n') | m' \neq m, n' \neq n, (m',n') \in X_{\text{nsf}}\} \\ S_3 &= \{(m,n') | n' \neq n, (m,n') \in X_{\text{nsf}}\} \\ S_4 &= \{(m',n) | m' \neq m, (m',n) \in X_{\text{nsf}}\}. \end{aligned}$$

It is obvious that $S_i, i = 1, 2, 3, 4$, is a division of X_{nsf} . Their cardinalities are

$$\begin{aligned} |S_1| &= 1 \\ |S_2| &= N^2 - 2(K+1)N + (K+1)^2 \\ |S_3| &= |S_4| = N - (K+1). \end{aligned}$$

Firstly, discarding the term $(m',n') \in S_1$ does not impact the limitation as it is of $O(1/N^2)$. Moreover, for $(m',n') \notin S_1$,

$$E[(1-x_{m,n})(1-x_{m',n'})] = (1-q)^2.$$

Leveraging the symmetry of entries in S_3 and S_4 , we obtain

$$\begin{aligned} E[A_N^2] &\rightarrow 1 - 2(1-q^2)^K \\ &+ \frac{1}{|X_{\text{nsf}}|} \left[\sum_{(m',n') \in S_2} + 2 \sum_{(m',n') \in S_3} \right] (E_{m',n'})^K. \end{aligned}$$

Since for $(m',n') \in S_2$, $(1-x_{m,j}x_{i,n})$ and $(1-x_{m',j}x_{i,n'})$ are independent, we have

$$\begin{aligned} E_{m',n'} &= E[1-x_{m,j}x_{i,n}] E[1-x_{m',j}x_{i,n'}] \\ &= (1-q^2)^2. \end{aligned}$$

For $(m',n') \in S_3$, we have

$$\begin{aligned} E_{m',n'} &= E[(1-x_{m,j}x_{i,n})(1-x_{m,j}x_{i,n'})] \\ &= E[1-x_{m,j}x_{i,n} - x_{m,j}x_{i,n'} + x_{m,j}x_{i,n}x_{i,n'}] \\ &= 1 - 2q^2 + q^3. \end{aligned}$$

Utilizing the facts:

$$\begin{aligned} \frac{|S_2|}{|X_{\text{nsf}}|} &= \frac{N^2 - 2(K+1)N + (K+1)^2}{N^2 - 2KN + K^2} \\ &\rightarrow 1 - \frac{2}{N} + o\left(\frac{1}{N}\right) \end{aligned}$$

$$\frac{|S_3|}{|X_{\text{nsf}}|} = \frac{N - (K+1)}{N^2 - 2KN + K^2} \rightarrow \frac{1}{N} + o\left(\frac{1}{N}\right)$$

And through some mathematical manipulations, we obtain

$$\begin{aligned} E[A_N^2] &\rightarrow \left(1 - (1-q^2)^K\right)^2 + \frac{2}{N} \left((1-2q^2+q^3)^K - (1-q^2)^{2K}\right). \end{aligned}$$

Here we have omitted higher-order terms of $o(1/N)$. Therefore,

$$\begin{aligned} D[A_N] &= E[A_N^2] - E[A_N]^2 \\ &\rightarrow \frac{2}{N} \left((1-2q^2+q^3)^K - (1-q^2)^{2K}\right) + o\left(\frac{1}{N}\right). \end{aligned}$$

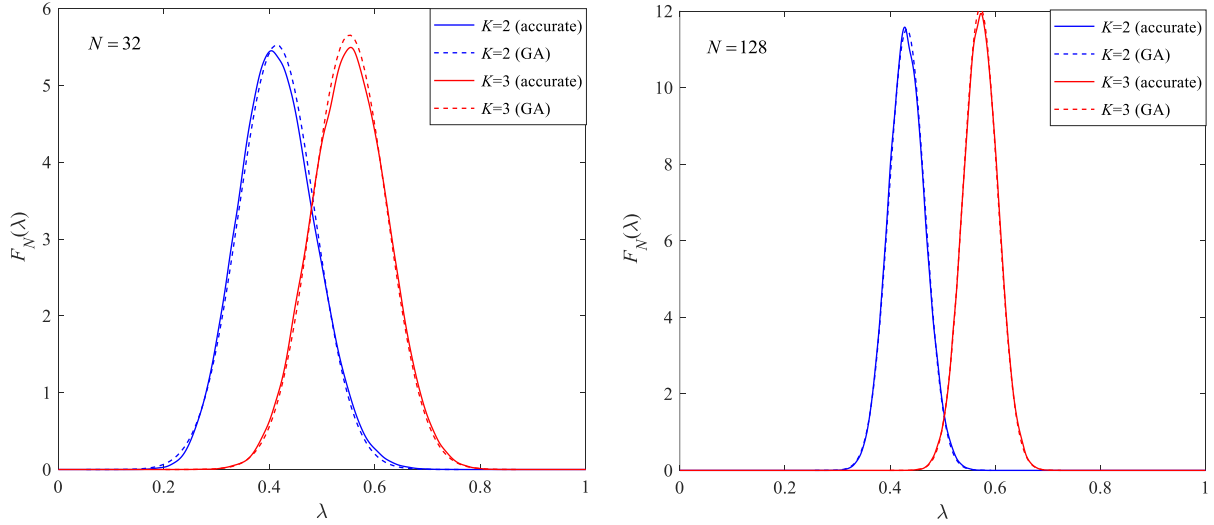


Fig. 6. Accurate probability distribution $F_N(\lambda)$ of SP rate and their Gaussian approximation (GA) when $q = 1/2$.

It leaves us to show that the remaining terms of (11) are all $o(1/N)$. Since $D[A_N] \rightarrow 0$ as $N \rightarrow \infty$, we have that $A_N \rightarrow E[A_N]$ in probability. Following the same procedure, we can show that $D[B_N] \rightarrow 0$ as $N \rightarrow \infty$ and thus, $B_N \rightarrow E[B_N]$ in probability. Since both A_N and B_N are bounded, i.e., $A_N \leq 1/(1-q)$, $B_N \leq 1/(1-q)$, we have $E[(A_N - E[A_N])(B_N - E[B_N])] \rightarrow 0$. Therefore, all the remaining terms of (11) are $o(1/N)$. Finally, we obtain

$$D[\lambda_N] \rightarrow \frac{2}{N} \left((1 - 2q^2 + q^3)^K - (1 - q^2)^{2K} \right) + o\left(\frac{1}{N}\right). \quad (12)$$

3) *Gaussian Approximation of $F_N(\lambda)$* : We establish that as $N \rightarrow \infty$, $F_N(\lambda)$ converges to a Gaussian PDF through the application of the strong Central Limit Theorem (CLT). This allows the Gaussian approximate for $F_N(\lambda)$

$$F_N(\lambda) \approx \frac{1}{\sqrt{2\pi D[\lambda_N]}} \exp\left(-\frac{(\lambda - E[\lambda_N])^2}{2D[\lambda_N]}\right) \quad (13)$$

where the first-order moment $E[\lambda_N]$ and variance $D[\lambda_N]$, derived in (10) and (12) respectively, are computed by neglecting $o(1/N)$ higher-order terms. Notably, Fig. 6 reveals that the Gaussian approximation remains highly accurate even for small array sizes, such as $N = 32$.

The strong CLT guarantees the asymptotic Gaussianity of λ_N . Through (8), we observe that λ_N is dominated by its first term $(1 - \frac{2K}{N})A_N$. We subsequently prove the Gaussian convergence of A_N as $N \rightarrow \infty$. Using the definitions

$$e_{m,n} = 1 - \prod_{(i,j) \in \Psi^*} (1 - x_{m,j}x_{i,n})$$

$$S_N = \left\{ (m,n) \mid (m,n) \in X_{\text{nsf}}, e_{m,n} = 1 \right\}$$

we reformulate A_N as

$$A_N = \frac{1}{|X_{\text{nsf}}|(1-q)} \sum_{(m,n) \in S_N} (1 - x_{m,n}).$$

Here, A_N constitutes a summation over a random number ($|S_N|$) of i.i.d. variables, where $0 \leq |S_N| \leq |X_{\text{nsf}}|$.

Following Rényi's strengthened CLT formulation [23], A_N converges to a Gaussian random variable provided that $\frac{|S_N|}{|X_{\text{nsf}}|} = \frac{1}{|X_{\text{nsf}}|} \sum_{(m,n) \in X_{\text{nsf}}} e_{m,n}$ converges in probability to a constant $c > 0$. We demonstrate that this ratio indeed converges in probability to its expectation $1 - (1 - q^2)^K$, employing a proof analogous to the convergence argument for A_N itself developed in previous subsections. This conclusively establishes the asymptotic Gaussianity of A_N . While the second term $\frac{2K}{N}B_N$ in (8) similarly exhibits Gaussian convergence, its influence on the overall distribution is negligible, so we omit the detailed analysis here.

B. Mutual Information and Dispersion of λ -Gaussian Channel

In this section, we analyze the mutual information I_λ and dispersion V_λ for the λ -Gaussian channel with a given input distribution. Given the conditions that $(R_0 - R'_0)/\sigma \gg 1$ and $\Delta \leq \lambda \leq 1 - \Delta$, where $\Delta > 0$ is a given small value, we propose approximate calculations for I_λ and V_λ . It is worth noting that the condition $(R_0 - R'_0)/\sigma \gg 1$ typically holds for the parameter settings provided in [10], [11], [12], and [13]. The condition $\Delta \leq \lambda \leq 1 - \Delta$ ensures that our analysis pertains to a non-trivial λ -Gaussian channel.

1) *Mutual Information I_λ* : Let $X \in \{0, 1\}$ and Y be the input and output, respectively, of the λ -Gaussian channel, as illustrated in Fig. 4. The conditional PDF of Y given X is expressed by

$$p_{Y|X}(y|x) = (1-x) \left(\lambda \phi(y - R'_0) + (1-\lambda) \phi(y - R_0) \right) + x \phi(y - R_1).$$

For input distribution $\Pr(X = 0) = 1 - q$ and $\Pr(X = 1) = q$, the marginal distribution of Y is

$$p_Y(y) = (1-q)p_{Y|X}(y|0) + qp_{Y|X}(y|1)$$

$$= (1-q) \left(\lambda \phi(y - R'_0) + (1-\lambda) \phi(y - R_0) \right) + q \phi(y - R_1).$$

Thus, we have

$$\begin{aligned}
 I(X; Y) &= E \left[\log \frac{P_{Y|X}(Y|X)}{P_Y(Y)} \right] \\
 &= qE \left[\log \frac{\phi(Z)}{P_Y(Z + R_1)} \right] \\
 &\quad + (1 - q)\lambda E \left[\log \frac{P_{Y|X}(Z + R'_0|0)}{P_Y(Z + R'_0)} \right] \\
 &\quad + (1 - q)(1 - \lambda)E \left[\log \frac{P_{Y|X}(Z + R_0|0)}{P_Y(Z + R_0)} \right] \quad (14)
 \end{aligned}$$

where $Z \sim \mathcal{N}(0, \sigma^2)$ is a Gaussian variable. Given the condition $(R_0 - R'_0)/\sigma \gg 1$ and $\Delta \leq \lambda \leq 1 - \Delta$, we provide approximation calculation for each term of (14). First, we have

$$\begin{aligned}
 &E \left[\log P_{Y|X}(Z + R'_0|0) \right] \\
 &= E \left[\log \left(\lambda \phi(Z) + (1 - \lambda)\phi(Z + R'_0 - R_0) \right) \right] \\
 &\leq E \left[\log \left(\lambda \phi(Z) \right) \right] + \frac{1}{\Delta} e^{-\frac{(R_0 - R'_0 - \beta)^2}{8\sigma^2}} \quad (15)
 \end{aligned}$$

where $\beta = \frac{2\sigma^2}{R_0 - R'_0} \log \frac{1 - \Delta}{\Delta}$, which is small when $(R_0 - R'_0)/\sigma$ is large. We provide the proof of (15) in Appendix A. Therefore, the first term of (15) is a good approximation for $E[\log P_{Y|X}(Z + R'_0|0)]$. Similarly, we have

$$\begin{aligned}
 &E \left[\log P_{Y|X}(Z + R_0|0) \right] \approx E \left[\log \left((1 - \lambda)\phi(Z) \right) \right] \\
 &E \left[\log P_Y(Z + R_1) \right] \\
 &\approx E \left[\log \left((1 - q)\lambda\phi(Z + R_1 - R'_0) + q\phi(Z) \right) \right] \\
 &E \left[\log P_Y(Z + R'_0) \right] \\
 &\approx E \left[\log \left((1 - q)\lambda\phi(Z) + q\phi(Z + R'_0 - R_1) \right) \right] \\
 &E \left[\log P_Y(Z + R_0) \right] \approx E \left[\log \left((1 - q)(1 - \lambda)\phi(Z) \right) \right].
 \end{aligned}$$

Note that the errors of these approximations are bounded by $\alpha e^{-((R_0 - R'_0)/2 - \beta)^2/(2\sigma^2)}$, where $\alpha > 0$. This can be strictly proven using the method presented in Appendix A.

After applying the approximations and performing some mathematical manipulations, we obtain

$$I(X; Y) \approx H(q') - (1 - q)H(\lambda) + q'I_{\text{BIAWGN}}\left(\frac{q}{q'}, \gamma\right) \triangleq I_\lambda \quad (16)$$

where $q' = q + (1 - q)\lambda$, $\gamma = \frac{R'_0 - R_1}{2\sigma}$, $H(x) = -x \log x - (1 - x) \log(1 - x)$ denotes the binary entropy function. $I_{\text{BIAWGN}}(q, \gamma)$ represents the mutual information of a BIAWGN channel with a Bernoulli (q) input and a signal-to-noise (SNR) of γ^2 , which is given by

$$\begin{aligned}
 I_{\text{BIAWGN}}(q, \gamma) &= -qE \left[\log \left(q + (1 - q)e^{2\gamma(Z - \gamma)} \right) \right] \\
 &\quad - (1 - q)E \left[\log \left(qe^{2\gamma(Z - \gamma)} + 1 - q \right) \right].
 \end{aligned}$$

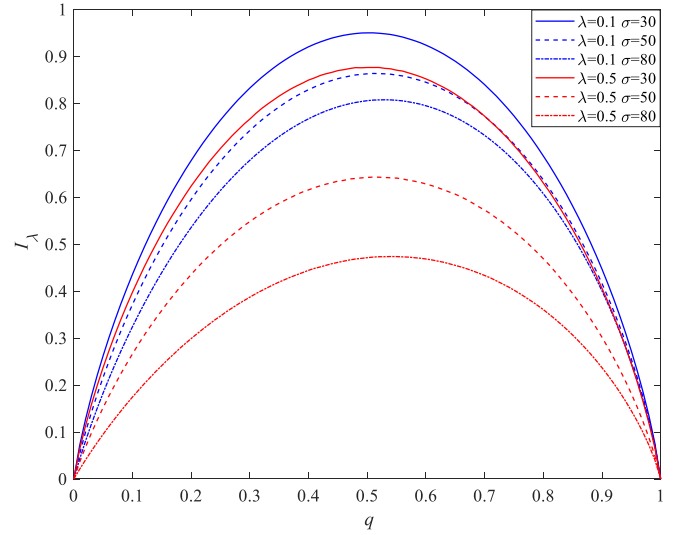


Fig. 7. Mutual information I_λ , $\lambda = 0.1, 0.5$, as functions of q for different channel noise levels $\sigma = 30, 50$, and 80 .

Interestingly, when $\lambda = 0$, we have $I_\lambda = H(q)$, which is the mutual information of a noiseless channel. This can be interpreted as that when $\lambda = 0$ and $(R_0 - R'_0)/\sigma \gg 1$ leading to $(R_0 - R_1)/\sigma \gg 1$, the channel becomes highly reliable. On the other hand, when $\lambda = 1$, we have $I_\lambda = I_{\text{BIAWGN}}(q, \gamma)$, which corresponds to the mutual information of the BIAWGN channel with an SNR of γ^2 . Therefore, the expression of I_λ remains valid in these two extreme cases.

It is worth mentioning that we can obtain an alternative form of I_λ using the following method. We define a Boolean variable $V \in \{0, 1\}$ as an indicator to determine whether an HRS occurs. Specifically, $V = 1$ if $Y = R_0 + Z$ and $V = 0$ otherwise. Thus, we have $\Pr(V = 0) = q'$ and $\Pr(V = 1) = 1 - q'$. When $V = 0$, Y takes the form of $Y = R_1 + Z$ with probability q/q' or $Y = R'_0 + Z$ with probability $1 - q/q'$. Furthermore, we can demonstrate that $H(Y) \approx H(Y, V)$, which subsequently implies $H(X|Y) \approx H(X|Y, V)$, using a technique similar to that employed in (15). The mutual information is then derived as

$$\begin{aligned}
 I(X; Y) &= H(X) - H(X|Y) \\
 &\approx H(X) - H(X|Y, V) \\
 &= H(X) - (1 - q')H(X|Y, V = 1) - q'H(X|Y, V = 0) \\
 &= H(X) - q'(H(X|V = 0) - I(X; Y|V = 0)) \\
 &= H(q) - q'H\left(\frac{q}{q'}\right) + q'I_{\text{BIAWGN}}\left(\frac{q}{q'}, \gamma\right). \quad (17)
 \end{aligned}$$

In (17) we have used the fact that given $V = 0$, X is a Bernoulli variable with $\Pr(X = 1) = q/q'$ and $\Pr(X = 0) = 1 - q/q'$. Additionally, given $V = 0$, the pair (X, Y) is an input and output of a BIAWGN channel. It can be verified that (17) and (16) are equivalent.

Figure 7 illustrates the mutual information I_λ ($\lambda = 0.1, 0.5$) as a function of q under various channel noise levels ($\sigma = 30, 50, 80$). Notably, despite the asymmetric nature of the λ -Gaussian channel, symmetrical input ($q = 0.5$) incurs almost no rate loss.

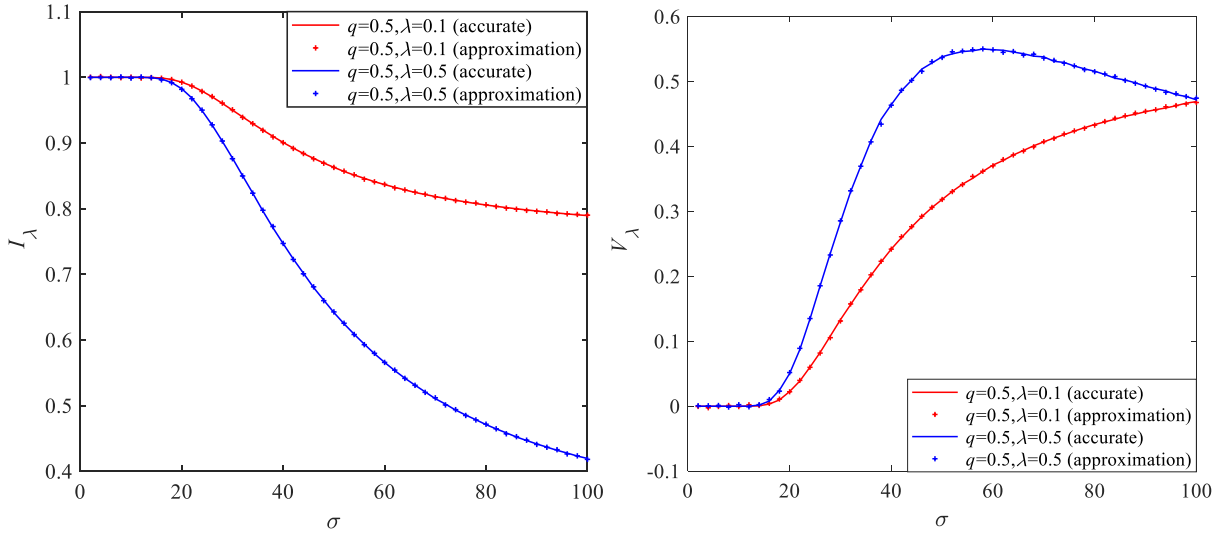


Fig. 8. Channel capacity and dispersion of λ -Gaussian channel: accurate values and approximations. The associated resistance values are $R_0 = 1000 \Omega$, $R_1 = 100 \Omega$, and $R_s = 250 \Omega$.

2) *Channel Dispersion V_λ* : The channel dispersion is defined as the variance of $\log \frac{P_{Y|X}(Y|X)}{P_Y(Y)}$. We utilize approximations similar to those presented in the previous sub-section, which are expressed in the form

$$E\left[\left(\log P_{Y|X}(Z + R'_0|0)\right)^2\right] \approx E\left[\left(\log(\lambda\phi(Z))\right)^2\right]. \quad (18)$$

The validity of this approximation is rigorously proven in Appendix B. It demonstrate that, for $(R_0 - R'_0)/\sigma \gg 1$ and $\Delta \leq \lambda \leq 1 - \Delta$, the error of this approximation is bounded by $\left(\frac{3(R_0 - R'_0)^4}{4\sigma^4} + \frac{3}{2\Delta^2}\right)e^{-\frac{(R_0 - R'_0 - \beta)^2}{8\sigma^2}}$, which is very small when $(R_0 - R'_0)/\sigma$ is large. Thus, we have

$$\begin{aligned} V_\lambda &= E\left[\left(\log \frac{P_{Y|X}(Y|X)}{P_Y(Y)}\right)^2\right] - I(X; Y)^2 \\ &\approx qE\left[\left(\log \frac{\phi(Z)}{(1-q)\lambda\phi(Z + R_1 - R'_0) + q\phi(Z)}\right)^2\right] \\ &\quad + (1-q)\lambda E\left[\left(\log \frac{\lambda\phi(Z)}{(1-q)\lambda\phi(Z) + q\phi(Z + R'_0 - R_1)}\right)^2\right] \\ &\quad + (1-q)(1-\lambda)(\log(1-q))^2 - (I_\lambda)^2 \\ &= qE\left[\left(\log\left((1-q)\lambda e^{2\gamma(\tilde{Z}-\gamma)} + q\right)\right)^2\right] \\ &\quad + (1-q)\lambda E\left[\left(\log\left(1 - q + \frac{q}{\lambda}e^{2\gamma(\tilde{Z}-\gamma)}\right)\right)^2\right] \\ &\quad + (1-q)(1-\lambda)(\log(1-q))^2 - (I_\lambda)^2 \end{aligned} \quad (19)$$

where $\tilde{Z} \sim \mathcal{N}(0, 1)$ is a Gaussian variable with mean 0 and unitary variance. Note that (19) remains valid when $\lambda = 1$ and $\lambda \rightarrow 0$.

Fig. 8 illustrates both the accurate and approximate mutual information I_λ and dispersion V_λ of λ -Gaussian channel for $q = 0.5$. Expectations in both the accurate and approximate

calculations are evaluated using Monte Carlo methods. With typical parameter settings of $R_0 = 1000 \Omega$, $R_1 = 100 \Omega$, and $R_s = 250 \Omega$, we observe almost no difference between the accurate and approximate values.

Ultimately, the approximate WEP bound of ReRAM channel can be obtained by substituting (13), (16), and (19) into (6).

IV. DENSITY EVOLUTION AND ITERATIVE DECODING THRESHOLD OVER λ -GAUSSIAN CHANNEL

In this section, we propose a DE framework for the design of practical sparse-graph codes over a λ -Gaussian channel. Sparse-graph codes, including LDPC codes and repeat-accumulate (RA) codes, have been proven to be asymptotic capacity-approaching [24]. A prominent feature of these codes is that their decoding can be efficiently implemented on a sparse factor graph using the belief propagation (BP) algorithm. By leveraging the iterative nature of BP, DE is commonly employed for predicting decoding performance and optimizing the codes. By tracking the PDF of the log-likelihood ratio (LLR) message, a decoding threshold can be determined under the assumption of infinite code length. We demonstrate the Gaussianity of the LLR messages over the λ -Gaussian channel. Based on this, we develop a low-complexity two-dimensional DE framework for code optimization. The analysis applies to a general sparse-graph code.

A. Gaussian Approximation of LLR

In this section, we demonstrate that the initial LLR provided to the decoder, as derived in (4), can be approximated as Gaussian. Furthermore, when $\lambda \approx 0$ and 1, the channel can be well approximated by a noiseless channel and a BIAWGN channel with an SNR of γ^2 , respectively. In these cases, the LLR exhibits a Gaussian distribution. Therefore, our subsequent analysis will concentrate on the scenario where $\Delta \leq \lambda \leq 1 - \Delta$, demonstrating the Gaussian nature of the LLR.

Since the λ -Gaussian channel produce three distinct types of outputs, specifically $Y = r + Z$, where r can be R_0, R_1 , and R'_0 , corresponding to the readback signals from the HRS, LRS, and SP cells of ReRAM, respectively, the LLR exhibits three different PDFs. We will individually establish the Gaussian nature of these PDFs.

1) *HRS Cell*: For $Y = R_0 + Z$ with $Z \sim \mathcal{N}(0, \sigma^2)$, the channel input should be $X = 0$. In this scenario, Y can be considered as the readback signal of an HRS cell in ReRAM. The LLR message (4) can be reformulated as

$$\begin{aligned} L_{\text{HRS}} &= \log \frac{(1-q)(1-\lambda)\phi(Y-R_0)}{q\phi(Y-R_1)} \\ &\quad + \log \left(1 + \frac{\lambda\phi(Y-R'_0)}{(1-\lambda)\phi(Y-R_0)} \right) \\ &= \frac{(R_0 - R_1)(R_0 - R_1 + 2Z)}{2\sigma^2} - \log \frac{q}{(1-q)(1-\lambda)} \\ &\quad + \log \left(1 + \frac{\lambda}{1-\lambda} e^{-\frac{(R_0-R'_0)(R_0-R'_0+2Z)}{2\sigma^2}} \right) \\ &\approx \frac{(R_0 - R_1)(R_0 - R_1 + 2Z)}{2\sigma^2} - \log \frac{q}{(1-q)(1-\lambda)} \quad (20) \end{aligned}$$

where we have neglected the last term of (20) with the assumption of $(R_0 - R'_0)/\sigma \gg 1$. As shown in Appendix A, the expected value of the absolute error of this approximation is in the form of $\alpha e^{-\frac{(R_0-R'_0-\beta)^2}{8\sigma^2}}$ where $\alpha > 0$ and $\beta = \frac{2\sigma^2}{R_0-R'_0} \log \frac{1-\Delta}{\Delta}$ for $\Delta < \lambda < 1 - \Delta$. Therefore, the LLR can be approximated as Gaussian

$$L_{\text{HRS}} \sim \mathcal{N} \left(\frac{(R_0 - R_1)^2}{2\sigma^2} - \log \frac{q}{(1-q)(1-\lambda)}, \frac{(R_0 - R_1)^2}{\sigma^2} \right).$$

Since $(R_0 - R'_0)/\sigma \gg 1$, the mean value of L_{HRS} is considerably large. Therefore, the readback signal of an HRS cell is quite reliable. During the DE process, we assume the message associated with an HRS cell is reliable.

2) *LRS Cell*: Similarly, for $Y = R_1 + Z$, the channel input should be $X = 1$. In this scenario, Y can be considered as the readback signal of an LRS cell in ReRAM. The LLR message (4) can be reformulated as

$$\begin{aligned} L_{\text{LRS}} &= \log \frac{\lambda(1-q)\phi(Y-R'_0)}{q\phi(Y-R_1)} \\ &\quad + \log \left(1 + \frac{(1-\lambda)\phi(Y-R_0)}{\lambda\phi(Y-R'_0)} \right) \\ &= -\frac{(R'_0 - R_1)(R'_0 - R_1 - 2Z)}{2\sigma^2} - \log \frac{q}{\lambda(1-q)} \\ &\quad + \log \left(1 + \frac{1-\lambda}{\lambda} e^{-\frac{(R'_0-R_0)(R'_0+R_0-2R_1-2Z)}{2\sigma^2}} \right) \\ &\approx -\frac{(R'_0 - R_1)(R'_0 - R_1 - 2Z)}{2\sigma^2} - \log \frac{q}{\lambda(1-q)}. \quad (21) \end{aligned}$$

Therefore, the LLR can be approximated as Gaussian

$$L_{\text{LRS}} \sim \mathcal{N} \left(-\frac{(R'_0 - R_1)^2}{2\sigma^2} - \log \frac{q}{\lambda(1-q)}, \frac{(R'_0 - R_1)^2}{\sigma^2} \right).$$

3) *SP Cell*: For $Y = R'_0 + Z$, the channel input should be $X = 0$. In this scenario, Y can be considered as the readback

signal of an SP cell in ReRAM. We follow a similar procedure and obtain

$$L_{\text{SP}} \approx \frac{(R'_0 - R_1)(R'_0 - R_1 + 2Z)}{2\sigma^2} - \log \frac{q}{\lambda(1-q)}.$$

Thus, we have

$$L_{\text{SP}} \sim \mathcal{N} \left(\frac{(R'_0 - R_1)^2}{2\sigma^2} - \log \frac{q}{\lambda(1-q)}, \frac{(R'_0 - R_1)^2}{\sigma^2} \right).$$

For all three cases, the commonly used assumption that the variance of LLR is equal to twice its mean is also approximately satisfied.

We compared the PDFs of $L_{\text{HRS}}, L_{\text{LRS}}$ and L_{SP} with their Gaussian approximation in Fig. 9 for $\lambda = 1/2$ and $q = 1/2$. It is evident that the Gaussian approximation is quite accurate. Furthermore, with typical parameter settings of $R_0 = 1000 \Omega, R_1 = 100 \Omega$, and $R_s = 250 \Omega$, L_{HRS} remains quite reliable even at a high noise level of $\sigma = 50$.

B. Two-Dimensional Representation of Message PDF

Due to the message asymmetry, assuming the LLRs to be single-parameter Gaussian is unreasonable. Based on the analysis presented in the previous sub-section, it is safe to assume the magnitude of L_{HRS} as infinite, i.e., the initial LLR of the HRS cell is reliable. Since the probability of $r = R_0$ is

$$p_0 = (1-q)(1-\lambda)$$

with probability p_0 , the initial message to the decoder is reliable. The reliable message serves as side information during iterative decoding, and does not require updating.

LLRs of an LRS and an SP cells exhibit similar magnitudes, differing only by the term $\pm \log \frac{q}{\lambda(1-q)}$. To see the magnitude of the LLR of L_{LRS} , we should normalize it as $-L_{\text{LRS}}$, since the message is associated with data $X = 1$. We approximate the LLRs of both LRS and SP cells using a single Gaussian variable with mean of

$$m_0 = \frac{(R'_0 - R_1)^2}{2\sigma^2} + \frac{q - \lambda(1-q)}{q + \lambda(1-q)} \log \frac{q}{\lambda(1-q)}.$$

This mean m_0 is a weighted average of the means of $-L_{\text{LRS}}$ and L_{SP} , with weights corresponding to their fractions $\frac{q}{q + \lambda(1-q)}$ and $\frac{\lambda(1-q)}{q + \lambda(1-q)}$. Specifically, we assume that the initial LLRs of both LRS and SP cells follow a Gaussian distribution $\mathcal{N}(m_0, 2m_0)$.

Based on the above analysis, we utilize the tuple (p, m) to specify the PDF of a message employed in decoding over a λ -Gaussian channel. Specifically, p represents the probability that the associated coded bit has already been recovered. Conversely, with probability $1 - p$, the coded bit remains unrecovered, and its LLR follows a Gaussian distribution of $\mathcal{N}(m, 2m)$. Thus, the tuple (p, m) fully captures the statistical characteristics of the message.

C. Density Evolution and Decoding Threshold

The decoder employs a standard BP decoding process on a sparse factor graph to recover the data. The graph comprises two types of nodes: variable and check nodes. The decoding is accomplished through local maximum a posteriori (MAP)

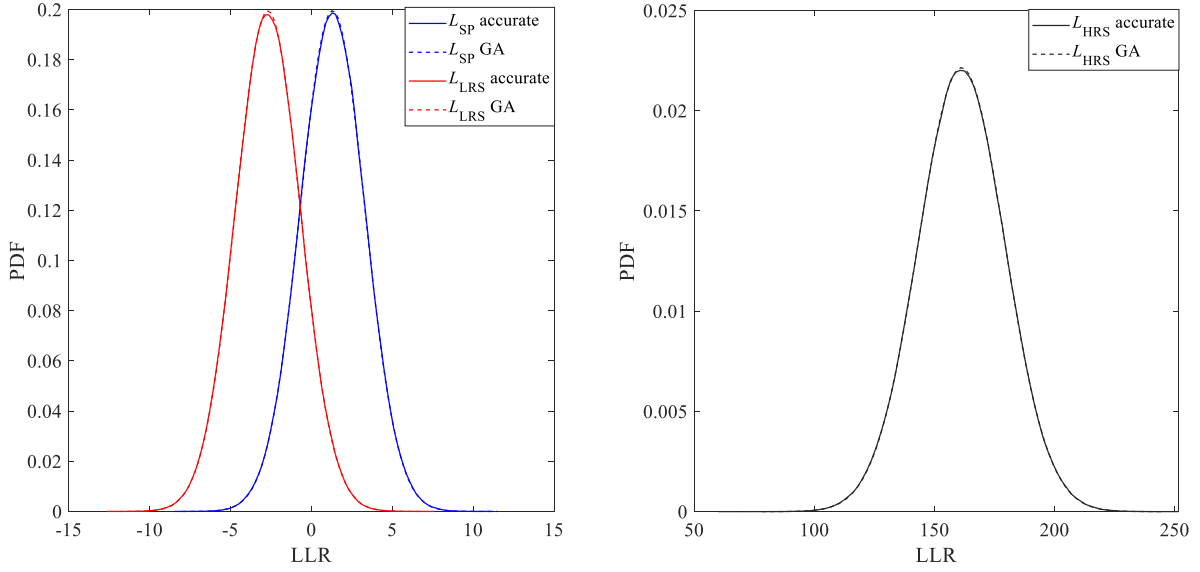


Fig. 9. Accurate PDFs of L_{LRS} , L_{SP} , L_{HRS} and their Gaussian approximation (GA) for $\lambda = 1/2$ and $q = 1/2$. The associated resistance values are $R_0 = 1000 \Omega$, $R_1 = 100 \Omega$, and $R_s = 250 \Omega$, and the noise standard deviation is $\sigma = 50$.

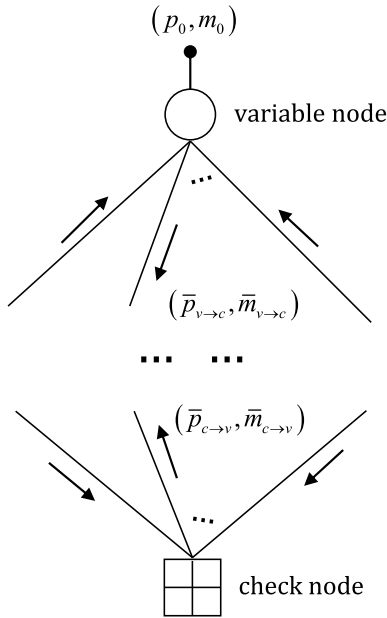


Fig. 10. Message passing over factor graph.

processing conducted by both the variable and check nodes, facilitated by the propagation of messages between them. Fig. 10 presents a general factor graph along with a schematic diagram illustrating the message passing. To track the PDF tuple of the message, we explore the update rules of both variable and check nodes. These rules are used to analyze the iterative decoding error probability and determine the decoding threshold. Our analysis is directed towards a general factor graph, thereby rendering it applicable to a wide range of sparse-graph codes.

1) *Variable Node Updating*: During iterative decoding, a variable node updates its output based on the incoming

message from the check nodes in its neighborhood. According to the standard BP, a degree- d variable node updates its message based on $d - 1$ LLR messages from check nodes and one initial LLR from the channel receive. The processing is a MAP estimation of a variable based on d independent observations. Specifically, let $(\bar{p}_{c \rightarrow v}, \bar{m}_{c \rightarrow v})$ be the PDF tuple for message from the check nodes. As presented in the previous sub-section, the PDF tuple of message from the λ -Gaussian channel is (p_0, m_0) . Thus, the PDF tuple (p_v, m_v) of a degree- d variable node output is derived as

$$\begin{aligned} p_v &= 1 - (1 - \bar{p}_{c \rightarrow v})^{d-1} (1 - p_0) \\ m_v &= m_0 + (d - 1) \bar{m}_{c \rightarrow v}. \end{aligned} \quad (22)$$

Here, (22) is derived based on the fact that the output of a variable node is reliable if and only if at least one of its inputs is reliable. Let $\{a_d, d = 1, 2, \dots, d_v\}$ represent the degree distribution of variable nodes, where a_d denotes the proportion of edges incident to variable nodes of degree d . The PDF tuple for the messages sent by the variable nodes after being averaged concerning its degree distribution is

$$\begin{aligned} \bar{p}_{v \rightarrow c} &= 1 - (1 - p_0) \sum_{d=1}^{d_v} a_d (1 - \bar{p}_{c \rightarrow v})^{d-1} \\ \bar{m}_{v \rightarrow c} &= m_0 + \left(\sum_{d=1}^{d_v} a_d d - 1 \right) \bar{m}_{c \rightarrow v}. \end{aligned}$$

2) *Check Node Updating*: A degree- d check node updates its output based on $d - 1$ incoming LLR messages from the variable nodes in its neighborhood. This operation follows the same rule as a MAP decoding of a single parity check code. Since a check node will output a reliable message if and only if all of its $d - 1$ incoming messages are reliable, we have

$$p_c = (\bar{p}_{v \rightarrow c})^{d-1}.$$

If i (where $0 < i \leq d-1$) incoming messages are unreliable, the output will be unreliable. In this case, according to the check node update rule [25], the mean of its output is

$$f(i) \triangleq \Phi^{-1} \left(1 - (1 - \Phi(\bar{m}_{v \rightarrow c})^i) \right) \quad (23)$$

where

$$\Phi(x) = \begin{cases} 1 - \frac{1}{\sqrt{4\pi x}} \int \tanh\left(\frac{u}{2}\right) e^{-\frac{(u-x)^2}{4x}} du & x > 0 \\ 1 & x = 0 \end{cases} \quad (24)$$

Since i follows a binomial distribution, we can obtain the mean output of the check node as

$$m_c = \sum_{i=1}^{d-1} \binom{d-1}{i} (\bar{p}_{v \rightarrow c})^{d-1-i} (1 - \bar{p}_{v \rightarrow c})^i f(i).$$

By averaging the check node output (p_c, m_c) concerning the degree distribution $\{b_d, d = 1, 2, \dots, d_c\}$, we obtain

$$\begin{aligned} \bar{p}_{c \rightarrow v} &= \sum_{d=1}^{d_c} b_d (\bar{p}_{v \rightarrow c})^{d-1} \\ \bar{m}_{c \rightarrow v} &= \sum_{d=1}^{d_c} b_d \sum_{i=1}^{d-1} \binom{d-1}{i} (\bar{p}_{v \rightarrow c})^{d-1-i} (1 - \bar{p}_{v \rightarrow c})^i f(i). \end{aligned}$$

Here, b_d represent the fraction of edges incident to check nodes of degree d .

3) *Decoding Threshold σ_{th}* : After a sufficient number of iterations, the two-dimensional PDF of LLR message output on each edge will converge to a fixed point, denoted as (p_∞, m_∞) . Thus, the error probability after a hard decision is derived as

$$P_e = (1 - p_\infty) Q \left(\sqrt{\frac{m_\infty}{2}} \right).$$

Note that, given λ and the degree distributions $\{a_d\}$ and $\{b_d\}$, P_e is solely determined by the channel noise parameter σ . The decoding threshold, denoted by σ_{th} , is the maximum tolerable value of σ that makes $P_e = 0$.

V. CODE OPTIMIZATION AND NUMERICAL RESULTS

This section presents a systematic code construction methodology grounded in our channel decomposition framework. While the proposed technique is universally applicable to sparse-graph code design, we select irregular repeat-accumulate (IRA) codes as a representative candidate due to their capacity-achieving properties under low-complexity BP decoding [26]. Fig. 11 illustrates the code's factor graph architecture, characterized by:

- (i) Systematic encoding structure
- (ii) Irregular variable node degree profile
- (iii) Uniform check node degree configuration.

The code optimization process focuses on joint refinement of variable-check node degree distributions through DE. Notably, conventional linear channel encoders inherently generate symmetric codeword distributions ($q = 0.5$), the design focus of this work. For asymmetric signaling schemes ($q \neq 0.5$), our framework requires integration with nonlinear distribution shaping mappers [13], [27], though this extension lies beyond our current scope.

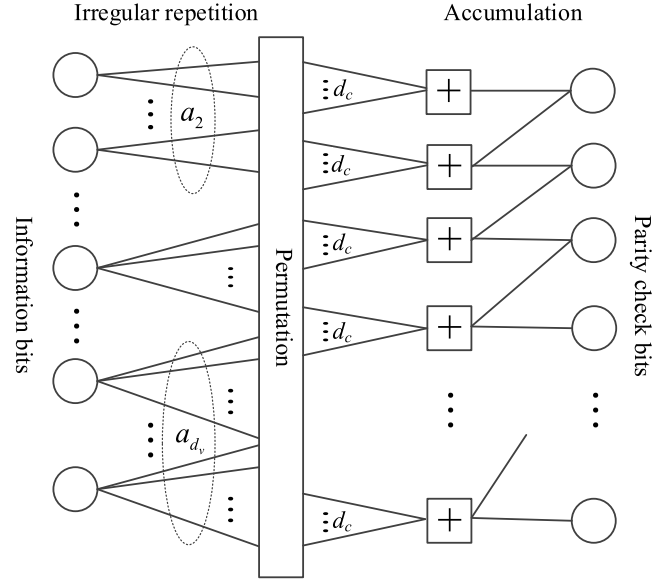


Fig. 11. Factor graph of IRA code.

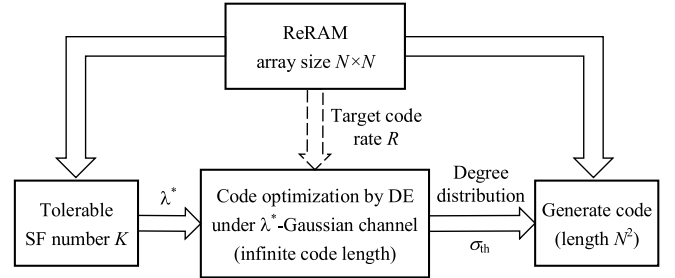


Fig. 12. Code design procedure for ReRAM channels.

Suppose we design codes for ReRAM arrays with size $N \times N$, where $N = 128$. To ensure that the code can tolerate up to K SFs, we will design the code under the λ^* -Gaussian channel, where $\lambda^* = E[\lambda_N] + 3\sqrt{D[\lambda_N]}$. Here, $E[\lambda_N]$ and $D[\lambda_N]$ are computed using (10) and (12) by neglecting $o(1/N)$ higher-order terms. This design approach enables the code to address the SP rates for $\Pr(\lambda_N < \lambda^*) \approx 1 - Q\left(\frac{\lambda^* - E[\lambda_N]}{\sqrt{D[\lambda_N]}}\right) \approx 99.87\%$ of the data patterns. The code design leverages our proposed two-dimensional DE framework, detailed in Sec. IV. The code optimization follows the following procedure. We utilize standard parameter values of $R_0 = 1000 \Omega$, $R_1 = 100 \Omega$, and $R_s = 250 \Omega$. We consider a constant check node degree of d_c and 3 types of variable node degrees of 3, 10, and 36. Then we employ a brute-force search to identify the optimal variable node degree distribution $\{a_3, a_{10}, a_{36}\}$ that achieves the target code rate R and also possesses the optimal decoding threshold σ_{th} . The code design procedure for ReRAM channels is outlined in Fig. 12. Following this procedure, we have obtained the IRA codes listed in TABLE I that perform very close to the information-theoretic bound σ^* . This critical parameter σ^* corresponds to the maximum tolerable noise level in the λ^* -Gaussian channel where mutual information satisfies $I_{\lambda^*} = R$.

Simulation results for the bit error rate (BER) and WER of the codes listed in TABLE I are illustrated in Figs. 13 and 14, respectively. These codes are simulated over the ReRAM

TABLE I

RATE-R IRA CODES TAILORED FOR 128×128 RERAM ARRAYS WITH K SFs IN THE ARRAYS. THE CODES ARE DESIGNED UNDER THE λ^* -GAUSSIAN CHANNEL. PARAMETER λ^* CORRESPONDS TO THE MAXIMUM TOLERABLE NOISE LEVEL IN THE λ^* -GAUSSIAN CHANNEL WHERE MUTUAL INFORMATION SATISFIES $I_{\lambda^*} = R$

$R = 0.5$							
K	λ^*	d_c	a_3	a_{10}	a_{36}	σ_{th}	σ^*
2	0.5338	6	0.3561	0.4165	0.2274	65	66
5	0.8306	6	0.3704	0.3560	0.2736	50	52
$R = 0.8$							
K	λ^*	d_c	a_3	a_{10}	a_{36}	σ_{th}	σ^*
1	0.3398	16	0.6540	0.3100	0.0360	37	39
2	0.5338	16	0.6878	0.1670	0.1452	33	35

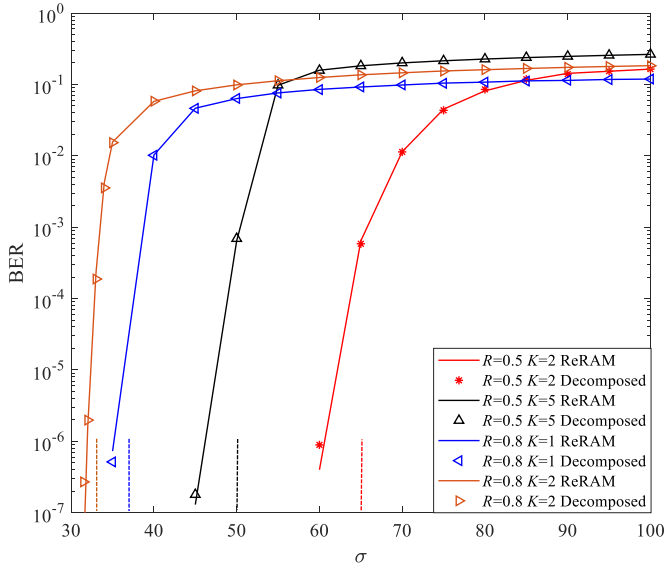


Fig. 13. BER of the codes listed in TABLE I over both ReRAM and the decomposed channels. The dashed lines illustrate the corresponding decoding thresholds over the λ^* -Gaussian channel.

channel configured with an $N \times N$ memory array with $N = 128$, yielding a code length $N^2 = 16384$. To address the error floor phenomenon in iterative decoding, the permutations in the IRA encoders were specifically designed to eliminate short cycles among degree-3 variable nodes in the factor graph. Both BER and WER metrics represent statistical averages computed across all transmitted codewords. Note that codes designed targeting K SFs can still work when the actual number of SFs in the array is fewer than K . However, to compare the simulation results with the corresponding decoding thresholds, we have simulated the code for ReRAM with exactly K active SFs. In each simulation, the locations of the SFs within the array are randomized. The standard BP decoding with 100 iterations is employed, with the channel parameter λ being estimated using (5) in each run. To validate our channel decomposition approach, we have also conducted simulations under the corresponding decomposed channel model, which is a mixed λ -Gaussian channel characterized by a mixing proportion of $F_N(\lambda)$.

In both figures, the simulation results obtained over the ReRAM channel coincide with those over the mixed λ -Gaussian channel, thereby confirming the validity of the

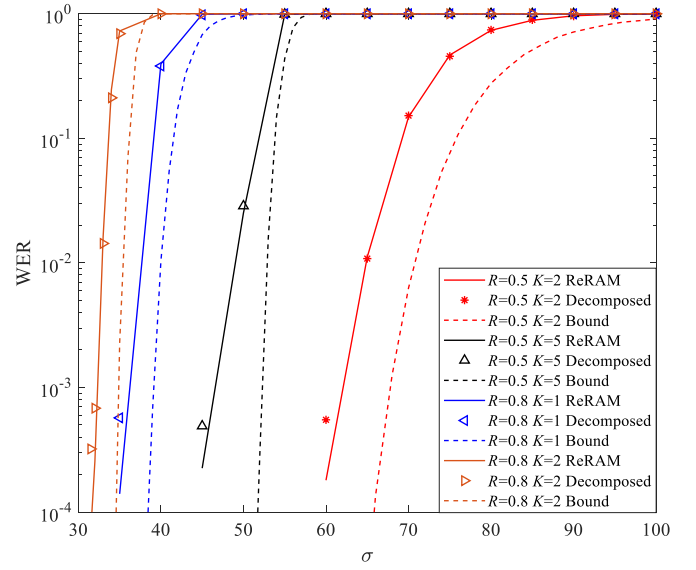


Fig. 14. WER of the codes listed in TABLE I over both ReRAM and the decomposed channels.

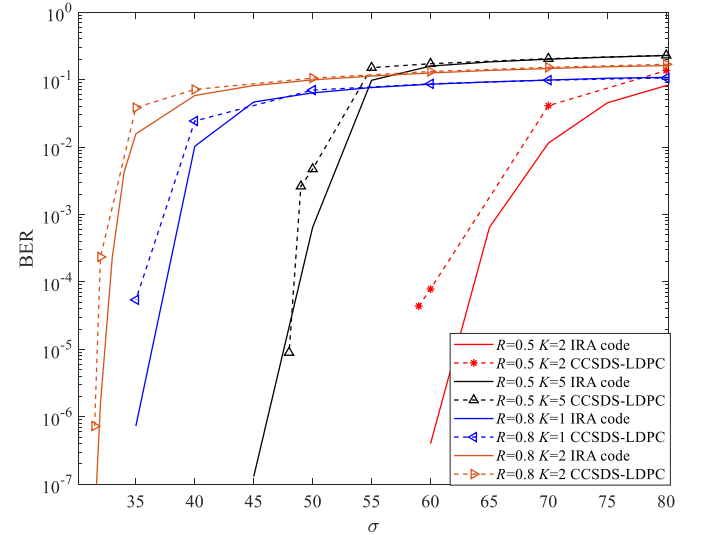


Fig. 15. The BER comparison between the IRA code designed in this work and the CCSDS standard LDPC code under the ReRAM channel.

channel decomposition approach. In Fig. 13, the gaps between the BER curves (at 10^{-6}) and the decoding thresholds σ_{th} over the corresponding λ^* -Gaussian channel are relatively small. In Fig. 14, we compared the WERs of our code with the finite-length WEP bound given by (6). It appears that there is a narrow gap between the WER curves and their respective bounds.

Finally, in Fig. 15 we compare the BER performance of the IRA code designed in this paper with Consultative Committee for Space Data Systems (CCSDS) standard LDPC codes of equivalent code rates over the ReRAM channel. It is noteworthy that this LDPC code demonstrates exceptional performance for AWGN channels (within 0.5 dB of the finite-length performance bound at a code length of 16384). Identical decoding iteration numbers were employed in simulations. Figure 15 reveals that for most channel conditions, the IRA codes designed in this paper outperform the CCSDS-LDPC codes in

the waterfall region owing to DE-based degree optimization.¹ At $R = 0.5$ and $K = 5$, the LDPC code exhibits a lower error floor, primarily due to the more sophisticated optimizations, such as cycle elimination and trapping set control, applied to the parity-check matrix of the CCSDS-standardized LDPC codes. This result confirms the effectiveness of the method proposed in this paper for ReRAM channel code design.

VI. CONCLUSION

We proposed a channel decomposition approach for performance analysis and channel code design over ReRAM channels. This approach decomposes the non-ergodic and data-dependent ReRAM channel into multiple stationary memoryless channels. By analyzing the mutual information and dispersion of these stationary memoryless channels, we derived an approximate finite-length performance bound. Leveraging this decomposition, we proposed a practical sparse-graph code design that utilizes the newly developed two-dimensional DE. Taking the IRA code as an illustrative example, we demonstrated our numerical results for code design. The designed codes not only exhibit capacity-approaching decoding thresholds but also yield WER simulation results that are close to the derived performance bound under practical crossbar configurations. Furthermore, simulations were conducted to confirm the equivalence between the original ReRAM channel and the decomposed channels.

APPENDIX

A. Proof of (15)

Denote $\Delta R = R_0 - R'_0$. For $\lambda \geq \Delta > 0$, we have

$$\begin{aligned} & E \left[\log \left(\lambda \phi(Z) + (1 - \lambda) \phi(Z + R'_0 - R_0) \right) \right] \\ &= E \left[\log \left(\lambda \phi(Z) \right) \right] + E \left[\log \left(1 + \frac{1 - \lambda}{\lambda} e^{-\frac{\Delta R(\Delta R - 2Z)}{2\sigma^2}} \right) \right] \end{aligned}$$

where $Z \sim \mathcal{N}(0, \sigma^2)$. Thus, we only need to show that the second term of the above expression is upper bounded by $\frac{1}{\Delta} e^{-\frac{(\Delta R - \beta)^2}{8\sigma^2}}$ where $\beta = \frac{2\sigma^2}{\Delta R} \log \frac{1 - \Delta}{\Delta}$.

Let $\tau = \frac{\Delta R}{2} + \frac{\sigma^2}{\Delta R} \log \frac{\lambda}{1 - \lambda}$. Since $\Delta R/\sigma \gg 1$, it is safe to assume $\Delta R > \tau > 0$. We have

$$\frac{1 - \lambda}{\lambda} e^{-\frac{\Delta R(\Delta R - 2Z)}{2\sigma^2}} > 1 \iff Z > \tau.$$

Utilizing the inequalities

$$\begin{aligned} \log(1 + x) &\leq x \quad \text{for } x > 0 \\ \log(1 + x) &\leq 1 + \log x \quad \text{for } x > 1 \end{aligned}$$

we obtain

$$\begin{aligned} & E \left[\log \left(1 + \frac{1 - \lambda}{\lambda} e^{-\frac{\Delta R(\Delta R - 2Z)}{2\sigma^2}} \right) \right] \\ &\leq E \left[\frac{1 - \lambda}{\lambda} e^{-\frac{\Delta R(\Delta R - 2Z)}{2\sigma^2}} 1\{Z \leq \tau\} \right] \\ &\quad + E \left[\left(1 + \log \frac{1 - \lambda}{\lambda} - \frac{\Delta R(\Delta R - 2Z)}{2\sigma^2} \right) 1\{Z > \tau\} \right]. \quad (25) \end{aligned}$$

¹Code design using density evolution is generally limited to optimizing performance in the waterfall region.

The first term of the above expression is bounded by

$$\begin{aligned} E \left[\frac{1 - \lambda}{\lambda} e^{-\frac{\Delta R(\Delta R - 2Z)}{2\sigma^2}} 1\{Z \leq \tau\} \right] &= \frac{1 - \lambda}{\lambda} Q \left(\frac{\Delta R - \tau}{\sigma} \right) \\ &\leq \frac{1 - \lambda}{2\lambda} e^{-\frac{(\Delta R - \tau)^2}{2\sigma^2}}. \quad (26) \end{aligned}$$

Using the formulations

$$\begin{aligned} E[1\{Z > \tau\}] &= Q \left(\frac{\tau}{\sigma} \right) \leq \frac{1}{2} e^{-\frac{\tau^2}{2\sigma^2}} \\ E[1\{Z > \tau\}Z] &= \frac{\sigma}{\sqrt{2\pi}} e^{-\frac{\tau^2}{2\sigma^2}} \end{aligned} \quad (27)$$

the second term is upper bounded by

$$\begin{aligned} & \left(\frac{1}{2} + \frac{1}{2} \log \frac{1 - \lambda}{\lambda} - \frac{\Delta R^2}{4\sigma^2} + \frac{\Delta R}{\sigma\sqrt{2\pi}} \right) e^{-\frac{\tau^2}{2\sigma^2}} \\ &\leq \left(\frac{1}{2} + \frac{1}{2} \log \frac{1 - \lambda}{\lambda} + \frac{1}{2\pi} \right) e^{-\frac{\tau^2}{2\sigma^2}} \leq \frac{1 + \lambda}{2\lambda} e^{-\frac{\tau^2}{2\sigma^2}}. \quad (28) \end{aligned}$$

where we used $\log x < x$ and $\frac{1}{2\pi} < 1/2$. Applying (26) and (28) to (25), and using $\lambda > \Delta$ and $\min\{\Delta R - \tau, \tau\} \geq \frac{\Delta R}{2} - \frac{\sigma^2}{\Delta R} \log \frac{1 - \Delta}{\Delta}$, we obtain

$$\begin{aligned} E \left[\log \left(1 + \frac{1 - \lambda}{\lambda} e^{-\frac{\Delta R(\Delta R - 2Z)}{2\sigma^2}} \right) \right] &\leq \frac{1}{\Delta} e^{-\frac{(\frac{\Delta R}{2} - \frac{\sigma^2}{\Delta R} \log \frac{1 - \Delta}{\Delta})^2}{2\sigma^2}} \\ &= \frac{1}{\Delta} e^{-\frac{(\Delta R - \beta)^2}{8\sigma^2}}. \end{aligned}$$

B. Proof of (18)

Denote $\Delta R = R_0 - R'_0$. For $\lambda \geq \Delta > 0$, we have

$$\begin{aligned} & E \left[\left(\log \left(\lambda \phi(Z) + (1 - \lambda) \phi(Z + R'_0 - R_0) \right) \right)^2 \right] \\ &\leq E \left[\left(\log \left(\lambda \phi(Z) \right) \right)^2 \right] \\ &\quad + E \left[\left(\log \left(1 + \frac{1 - \lambda}{\lambda} e^{-\frac{\Delta R(\Delta R - 2Z)}{2\sigma^2}} \right) \right)^2 \right] \quad (29) \end{aligned}$$

due to $\log(\lambda \phi(Z)) \leq \log \frac{\lambda}{\sqrt{2\pi\sigma^2}} < 0$ which holds for typical values of σ we considered. We show that the second term of (29) is less than $\left(\frac{3\Delta R^4}{4\sigma^4} + \frac{3}{2\Delta^2} \right) e^{-\frac{(\Delta R - \beta)^2}{8\sigma^2}}$ where $\beta = \frac{2\sigma^2}{\Delta R} \log \frac{1 - \Delta}{\Delta}$.

Let $\tau = \frac{\Delta R}{2} + \frac{\sigma^2}{\Delta R} \log \frac{\lambda}{1 - \lambda}$. Using the similar technique as in Appendix A, we have

$$\begin{aligned} & E \left[\left(\log \left(1 + \frac{1 - \lambda}{\lambda} e^{-\frac{\Delta R(\Delta R - 2Z)}{2\sigma^2}} \right) \right)^2 \right] \\ &\leq E \left[\frac{(1 - \lambda)^2}{\lambda^2} e^{-\frac{\Delta R(\Delta R - 2Z)}{\sigma^2}} 1\{Z \leq \tau\} \right] \\ &\quad + E \left[\left(1 + \log \frac{1 - \lambda}{\lambda} - \frac{\Delta R(\Delta R - 2Z)}{2\sigma^2} \right)^2 1\{Z > \tau\} \right]. \quad (30) \end{aligned}$$

The first term of the above expression is bounded by

$$\begin{aligned} & E \left[\frac{(1 - \lambda)^2}{\lambda^2} e^{-\frac{\Delta R(\Delta R - 2Z)}{\sigma^2}} 1\{Z \leq \tau\} \right] \\ &= \frac{(1 - \lambda)^2}{\lambda^2} e^{-\frac{\Delta R^2}{\sigma^2}} Q \left(\frac{2\Delta R - \tau}{\sigma} \right) \end{aligned}$$

$$\begin{aligned}
&\leq \frac{(1-\lambda)^2}{2\lambda^2} e^{-\frac{(2\Delta R - \tau)^2 - 2\Delta R^2}{2\sigma^2}} \\
&\leq \frac{1}{2\Delta^2} e^{-\frac{\tau^2}{2\sigma^2}}
\end{aligned} \quad (31)$$

where we have used $\frac{(1-\lambda)^2}{2\lambda^2} \leq \frac{1}{2\Delta^2}$ and $\tau \leq \Delta R$.

To evaluate the second term of (30), we first have

$$\begin{aligned}
&\left(1 + \log \frac{1-\lambda}{\lambda} - \frac{\Delta R^2 - \Delta R Z}{2\sigma^2}\right)^2 \\
&\leq 2 \left(1 + \log \frac{1-\lambda}{\lambda}\right)^2 + 2 \left(\frac{\Delta R^2 - \Delta R Z}{2\sigma^2}\right)^2 \\
&\leq \frac{2}{\Delta^2} + \frac{\Delta R^4 + \Delta R^2 Z^2}{2\sigma^4}
\end{aligned} \quad (32)$$

where we have used $|1 + \log \frac{1-\lambda}{\lambda}| \leq \frac{1}{\Delta}$ and the condition $Z > \tau > 0$. Moreover, since $\tau < \Delta R$ and $\sigma < \Delta R$, we have

$$E[1\{Z > \tau\}Z^2] = \frac{\sigma\tau}{\sqrt{2\pi}} e^{-\frac{\tau^2}{2\sigma^2}} + \sigma^2 Q\left(\frac{\tau}{\sigma}\right) \leq \Delta R^2 e^{-\frac{\tau^2}{2\sigma^2}}. \quad (33)$$

Utilizing (27), (32), and (33), we obtain that the second term of (30) is upper bounded by

$$\begin{aligned}
&E\left[\left(1 + \log \frac{1-\lambda}{\lambda} - \frac{\Delta R(\Delta R - 2Z)}{2\sigma^2}\right)^2 1\{Z > \tau\}\right] \\
&\leq \left(\frac{1}{\Delta^2} + \frac{3\Delta R^4}{4\sigma^4}\right) e^{-\frac{\tau^2}{2\sigma^2}}.
\end{aligned} \quad (34)$$

Applying (31) and (28) to (30), we obtain

$$\begin{aligned}
&E\left[\left(\log\left(1 + \frac{1-\lambda}{\lambda} e^{-\frac{\Delta R(\Delta R - 2Z)}{2\sigma^2}}\right)\right)^2\right] \\
&\leq \left(\frac{3\Delta R^4}{4\sigma^4} + \frac{3}{2\Delta^2}\right) e^{-\frac{(\Delta R - \beta)^2}{8\sigma^2}}
\end{aligned}$$

where we have used $\tau \geq (\Delta R - \beta)/2$.

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Guanghui Song (Member, IEEE) received the B.E. degree in communication and engineering from Henan Normal University, Xinxiang, China, in 2006, the M.S. degree in telecommunications engineering from Xidian University, Xi'an, China, in 2009, and the Ph.D. degree in the Department of Intelligent Information Engineering and Sciences, Doshisha University, Kyoto, Japan, in 2012. From 2013 to 2021, he did Post-Doctoral Research with Doshisha University, Kyoto, the University of Western Ontario, London, Canada, and Singapore University of Technology, and Design, Singapore. He is currently an Associate Professor with Xidian University, Xi'an, China. His research interests include channel coding, multi-user coding, and coding for data storage systems.

Meiru Gao received the B.E. degree in communication engineering from North University of China, Taiyuan, China, in 2022, and the M.S. degree in telecommunications engineering from Xidian University, Xi'an, China, in 2025. She joined CMB Network Technology as a Systems Engineer. Her research interests include wireless communication and channel coding theory.

Ying Li (Member, IEEE) received the B.E. degree in telecommunication engineering and the Ph.D. degree in communication and information systems from Xidian University, Xian, China, in 1995 and 2005, respectively. From 2011 to 2012, she was with the University of California at Davis, Davis, CA, USA, as a Visiting Scholar. She is currently a Professor with Xidian University. Her current research interests include design and analysis for wireless systems, including channel coding, wireless network communications, interference processing, and MIMO techniques.

Bin Dai (Member, IEEE) received the Ph.D. degree from the School of Electronic and Information Engineering, Beihang University, Beijing, China, in 2019. From 2017 to 2018, he was a Visiting Scholar with Lehigh University, Bethlehem, PA, USA. From 2023 to 2025, he was a Post-Doctoral Researcher with Singapore University of Technology and Design, Singapore. He is currently an Associate Professor with the School of Internet of Things, Nanjing University of Posts and Telecommunications, Nanjing, China. His research interests include signal processing, coding theory, and deep learning with applications in data storage and wireless communications.

Kui Cai (Senior Member, IEEE) received the B.E. degree in information and control engineering from Shanghai Jiao Tong University, Shanghai, China, and joint Ph.D. degree in electrical engineering from the Technical University of Eindhoven, The Netherlands, and National University of Singapore. She is currently an Associate Professor with Singapore University of Technology and Design (SUTD). Her main research interests include coding theory, information theory, and signal processing for various data storage systems and digital communications. She received 2008 IEEE Communications Society Best Paper Award in coding and signal processing for data storage. She is served as the Vice-Chair (Academia) of IEEE Communications Society, Data Storage Technical Committee (DSTC) from 2015 and 2016.

Lin Zhou (Senior Member, IEEE) received the B.E., M.E., and Ph.D. degrees in information and communications engineering from Xidian University, Xi'an, China, in 2004, 2007, and 2011, respectively. He is currently a Tenured Associate Professor with the College of Information Science and Engineering, Huaqiao University, Xiamen, China. His research interests include channel coding, coded modulation, joint source-channel coding, intelligent information processing, and the industrial Internet. He has served as the Chair or TPC member for various IEEE conferences. He holds 16 invention patents in China, with nearly 20 more pending.